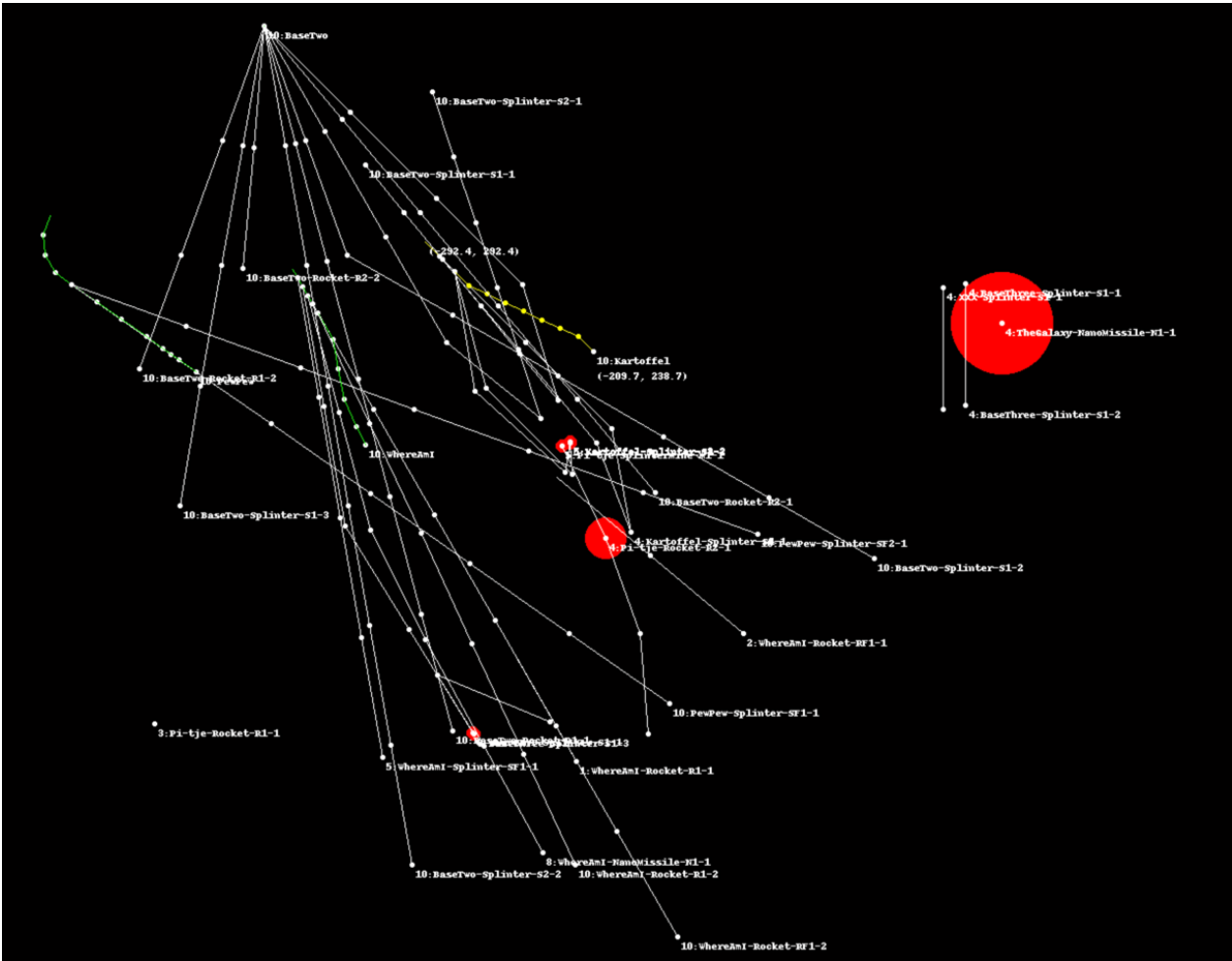


Welcome to Space Arena

Generated on 30 Jul 2026



A typical round of mayhem...

This game is a recreation of an old Play-By-Mail game from 1991. In this game teams of space ships battle each other in a 2D arena. The game is a simulation where players all send in their orders before a round is processed. A round consists of 10 clock *ticks* and the results of the players' commands gets processed into a result output (HTML & PDF). Each ship receives their own result based on what they observed. This information is used to plan the next round and send in new orders, and the cycle continues until there is a winner.

Game rounds

The game is played in rounds of 10 ticks. Each round you send in a set of commands for that round. When the round is processed, all ships move and act according to their commands. The results of the round are then sent back to each ship's player and the cycle continues until the end conditions are reached.

Scoring

Event	Score
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Damage shield	0.5 point per point of shield damaged
Break shield (to 0 strength)	25 points
Damage hull	2 points per point of hull damaged
Drain battery	0.5 points per point of energy drained
Deal killing blow to other ship (0 hull)	100 points

Commands

[Overview](#) | [Rd 0](#) | [Rd 1](#) | [Rd 2](#) | [Rd 3](#) | [Rd 4](#) | [Rd 5](#) | [Current round](#)

Kartoffel: plan next round

x: A(amount)

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x: R(degrees)

x: L(degrees)

x: Activation (component) (on/off)

— Ship Moves

x: Fire (missile launcher) (direction)

x: Fire (laser) (target)

x: Scan (scanner) (direction) (scan cone)

x: Replenish

x: Boost (N/S/E/W) (amount)

Status

Position (424.2, -304.9) **Heading** 135 |30| **Speed** 30/30 |15|
Hull 150/150 **Battery** 242/500.
Shields N (260) E (125) S (50) W (125)

L1 Firing Arc ((315, 45)) Temperature (0/100) Strength (150)
S1 Ammo (0 Splinter) Firing Arc ((315, 45))
S2 Ammo (0 Splinter) Firing Arc ((315, 45))
S3 Ammo (0 Splinter) Firing Arc ((315, 45))
S4 Ammo (0 Splinter) Firing Arc ((315, 45))
N1 Ammo (12 NanoMissile) Firing Arc ((315, 45))
R1 Ammo (16 Rocket) Firing Arc (360)
M1 Ammo (10 SplinterMine) Firing Arc (360)
G Firing Arc (360)

You can access your command interface by clicking on your ship's name, or when you're in a ship's turn by pressing "Current round" at the top of the page. Enter one command per line, with the format (tick number): (command). **You can write multiple commands per tick**, but they each should be on their own line. Whitespace in the commands does not matter: you can write R 90 or R90, whatever is your preference.

You can use the "Check" button for some (basic) checking, and the "Save" button to save your turn for processing. Note that as long as the turn is not processed you can save as many times as you want.

The following commands let the ship accelerate by 25 and turn 90 degrees left on tick 1, and fire a rocket from the launcher called 'L1' 90 degrees from its current heading on tick 2. Note that the A25 line is after the L90 line: the acceleration will always be processed before the turn in that tick.

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1: L90
1: A25
2: Fire L1 90

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Commands are for changes: you do not have to enter a command for every tick. If given no commands, your ship will fly straight at its current speed.

It does not matter in which order you put these lines, and it also does not matter if the tick numbers of the lines are out of order. All orders get processed in tick number order, and within a tick orders get processed in a specific order. Empty lines and lines starting with '#' are ignored. You can use this for instance to group your movement and weapon commands together.

You can fire each weapon once per tick, and they might have temperature or ammo to take into account. If you enter multiple commands to fire a specific weapon in the same tick, only the first command is executed. In this case line order *does* matter.

Command	Parameter(s)	Example	Purpose	Timing	Notes
A	+/- Integer	A - 30	Increase or decrease speed in units per tick	One per tick, pre-move, before turn command	Limited to maximum acceleration, can be negative.
R / L / H	+/- degrees	L 45	Turn right or left in degrees	One per tick, pre-move, after acceleration command	Limited to maximum turn rate per tick. However, at speed 0 there is no limit to turn rate.
F / Fire	[Weapon Name] [target or direction]	Fire L1 MasterBlaster F M1 90	Fire a named weapon. Depending on the weapon the next parameter is a direction (e.g. rockets) or the name of another object (e.g. Laser)	Post-move	Can fire each weapon once per tick
Replenish	No parameters	Replenish	Replenish the ship (ammo, hull, shields, temperature, etc.) when close to a star base.	Post-move	Only happens if after the move the ship is within 10 units and slower than 10 speed.
Boost	[Shield quadrant] [Amount]	Boost N 40	Boost a shield quadrant by amount. This costs 1 point of energy per point of shield.	Post-move	You can boost a shield quadrant to twice its normal maximum. However, at the end of a round any excess dissipates.
Activation	[Component] [On/Off]	Activation C1 On	Activate/deactivate a component, for components that need activation.	Pre-move	Currently the only component is a Cloak on the H2552 that reduces your visibility by 20% for 10 energy per tick.
Scan	[Component] [Direction] [Scan Cone]	Scan G 75 30	Long range scan, smaller cone means longer distance.	Post-move	The scan range is closest a scan cone of 360 degrees, furthest at 30 degrees. Smallest scan cone is 30 degrees.

A command set for a round could look like below. Note that the order of the lines can be mixed: there is a fixed order that commands will be executed.

8: A-30 8: L180 1: Fire M1 35 1: Fire M1 50 6: R30 7: R30 1: L20 1: A30 1: A40 8: Replenish 9: A30	• <i>Tick 1:</i> the ship accelerates by 70 (1: A30, 1: A40) but will likely be limited to its maximum acceleration. • <i>Tick 1:</i> the ship turns left by 20 degrees (1: L20) • <i>Tick 1</i> Tick 1: the ship fires a rocket 35 degrees to the right of its heading (1: Fire M1 35) • The second Fire command is ignored because it tries to fire the "M1" weapon for a second time that tick (1: Fire M1 50) • <i>Tick 6 and 7:</i> the ship turns right by 30 degrees for two ticks (6: R30, 7: R30) • <i>Tick 8:</i> the ship slows down by 30 (8: A-30) and because its speed is 0 it can make a full reverse turn (8: L180)
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- | | |
|--|--|
| | <ul style="list-style-type: none">• <i>Tick 8</i>: the ship - assuming it is close enough to a star base - fully replenishes (8: Replenish)• <i>Tick 9</i>: the ship accelerates by 30 to fly back to where it came from (9: A30) |
|--|--|

Tick Processing

Each tick is processed in a specific order, it is useful to know in what order everything is processed. For instance, acceleration is done before turning, so you can do a fast turn within one tick by slowing down to 0 and then turning as much as you want.

1. Start of tick. Some components do upkeep (Laser cools down).
2. Generators generate energy, 1 energy per generator.
3. Components that use energy per tick use their energy, draining the battery. Also the energy drain for the upcoming movement is deducted.
4. Pre-move commands are executed.
 1. Acceleration
 2. Turning
 3. Other pre-move commands (Activation)
4. Movement
5. Post-move commands are executed.
 1. Fire weapons. This is when new missiles appear. Damage is done.
 2. Boost
3. Ships, guided missiles etc. scan their environment.
4. Guided missiles choose and intercept their target. Missiles in range of target explode and do damage.
5. Destroyed objects are removed.

Energy

All the ships energy is stored on its central battery, and it can power several things, like lasers, shields and ship's movement. Every generator on a ship produces 1 energy per tick and stores it in the battery.

Energy is used to:

- Boost shields with the Boost command at 1 point of shield per point of energy.
- Power movement at Speed / 10 energy per tick.
- Power lasers at 5 energy per shot.

Don't forget: a Replenish **completely** fills the battery!

Movement

In movement everything is per tick.

Any object moves its speed every tick. So a ship with a speed of 30 moves $10 * 30 = 300$ units in a full round of 10 ticks. A ship has a maximum speed, and can fly backwards with the same (negative) speed.

An object has a maximum turn rate in degrees per tick. An object with a maximum turn rate of 30 can make a full circle in $360 / 30 = 12$ ticks.

An object has a maximum acceleration in speed units per tick. So an object with a maximum acceleration of 20 and a maximum speed of 40 can reach maximum velocity in 2 ticks.

Movement costs energy, at speed / 10 points energy per tick, rounded down. For instance, a ship with a speed of 46 will spend $46 / 10$ rounded down = 4 energy per tick on movement.

Movement commands

The movement commands are all *per tick*. They are:

A[number]
Accelerate or decelerate by x units that tick. You can fly backwards (negative speed)
L[degrees]
Turn x degrees to the left that tick.
R[degrees]
Turn x degrees to the right that tick.

All movement-related commands are **relative** commands. Turning x degrees with respect to current heading, acceleration changes current speed by x amount, etc.

At speed 0 there is no limit to the turn rate.

If you give multiple turn or accelerate commands *in a single tick*, they get combined (added) into one command for that tick. If you turn beyond the maximum turn rate of the ship, the command gets constrained to the maximum. The same goes for maximum acceleration and speed.

Weapons

Lasers

Lasers are fired at a specific ship and always hit if they are in range. Lasers heat up per shot and can't fire when they are at or over their maximum temperature. Lasers always heat and use energy when fired, even when there's no target, so don't just spam fire commands!

Energy per shot	Damage per shot	Heat per shot	Max Temperature
5	Strength - Distance	20	100

Missiles

Missiles are fired in a direction from a missile launcher. Guided missiles (Splinter) will target the nearest object they see, but not the firing ship or its missiles. Regular Rockets fly in a straight line. A missile has a triggering range. Once a target moves within this range, the missile will explode. When a missile explodes, it damages **everything** in range, including its origin ship and rockets. Guided missiles will scan for near targets in a 45 degree cone in front of them and adapt their trajectory to intercept. These missiles do not target their origin ship and its missiles.

Missile launchers have limited ammo, but can be restocked by replenishing at a star base. Currently all missiles have speed 60 and fizzle out after 15 ticks. This means they have a range of $60 * 15 = 750$.

Rockets are unguided missiles with a trigger- and blast radius of 20 and 50 damage. They fly at a speed of 60 units per tick.

Splinter missiles are guided missile that chase the closest target that is not its originator or its missiles. They fly at a speed of 60 units per tick. The missile will scan for targets in cone shaped 45 degrees and 150 distance units in front. It has a trigger- and blast radius of 6, and their damage linearly scales down from 75 at 0 distance to 0 damage at 6 distance. They are quite smart in that they predict a target's movement and will on approach will slow down to exactly close the distance to their target. Guided missiles will target other missiles, so shooting a missile back at them is a good way to defend against them.

Power Splinter missiles are more powerful versions of the Splinter missile. They have a base damage of 100 vs. the regular Splinter's 75.

Nano missiles release a cloud of nanites that are specialized in eating through matter, and are therefore dangerous to a ship's hull: they do double damage to it. However, they completely burn up against a shield and do not damage it, so this weapon will **only** work if it hits a ship through a broken shield!

EMP missiles drain energy from shields and batteries. When they hit shields they do double damage, and any remaining strength when it hits the hull does not damage it, but drains the battery. EMP missiles score 1 point for every two units of energy they drain, whether from shields or battery.

Scanners

Gravscan

The Grav scanner is a component that you can use to find opponents at long range. Each ping costs 10 energy, but it allows you to scan at long distances, up to 6000 units at the smallest scan cone (30 degrees).

Defense

Shields

Shields are the outer defense of a ship: a shield has to be broken before the hull is damaged. A ship's shield is divided in quadrants that each cover an arc of 90 degrees (North, East, South, West). Each quadrant has its own strength and needs to be boosted separately. Each point of shield prevents one point of damage.

Shields can be **Boosted** to recover them. This uses energy from the battery: 1 point of energy for 1 point of shield. Shields can be boosted to twice their normal maximum, but any excess strength is lost at the end of the round. However, well timed boosts allow you to tank quite some damage!

Cloak

Ships and guided missiles have a scan range within which they can scan objects around them. Some ships have a cloaking device. It shortens the range at which the cloaked ship can be detected. The strength of the cloak determines how much the others' scan range is shortened when detecting the cloaked ship.

If the strength of a cloak is 0.2, it makes the others' scan range a factor 0.2 (20%) shorter, e.g. $100 - 20\% = 80$. A ship with a scan range of 100 will not detect a strength-0.2-cloaked ship if it is beyond a range of 80.

Cloaks use 10 energy per tick, and can be turned on and off with the **Activate** command.

Ships

Ships are player-controlled war machines. They come in different configurations.

A2527 Alligator

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	35	25	100	60/500	7	210
Weapons				Defense		
L1: Laser (140 (360)) S1: Launcher (5 PowerSplinter (360)) N1: Launcher (5 NanoMissile (315, 45)) R1: Launcher (10 Rocket (360)) RF1: Launcher (12 Rocket (315, 45)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (100/100/100/100)		

A2539 Caiman

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	35	25	100	80/500	7	180
Weapons				Defense		
L1: Laser (110 (360)) LF1: Laser (110 (315, 45)) S1: Launcher (5 PowerSplinter (360)) SF1: Launcher (5 PowerSplinter (315, 45)) N1: Launcher (10 NanoMissile (360)) R1: Launcher (5 Rocket (360)) RF1: Launcher (5 Rocket (315, 45)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (100/100/100/100)		
ECM						
Cloak: Cloak (0.2)						

A2545 Terrapin

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	30	20	150	80/500	5	270
Weapons				Defense		
S1: Launcher (6 PowerSplinter (360)) S2: Launcher (6 PowerSplinter (360)) SF1: Launcher (6 PowerSplinter (270, 90)) N1: Launcher (12 NanoMissile (270, 90)) R1: Launcher (8 Rocket (360)) RF1: Launcher (12 Rocket (270, 90)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (110/110/110/110)		

A2553 Frog

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	40	20	80	70/500	6	240
Weapons				Defense		
L1: Laser (120 (360)) S1: Launcher (5 PowerSplinter (360)) SF1: Launcher (5 PowerSplinter (315, 45)) N1: Launcher (5 NanoMissile (360)) R1: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (100/100/100/100)		
ECM						
Cloak: Cloak (0.2)						

F2551 Tiger

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
50	50	30	75	90/500	8	180
Weapons				Defense		
L1: Laser (150 (360)) L2: Laser (150 (360)) R1: Launcher (10 Rocket (360)) R2: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (110/100/100/100)		
ECM						
Cloak: Cloak (0.25)						

F2547 Panther

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
50	45	25	80	90/500	8	180
Weapons				Defense		
L1: Laser (130 (360)) S1: Launcher (3 Splinter (270, 90)) N1: Launcher (3 NanoMissile (270, 90)) R1: Launcher (10 Rocket (360)) R2: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (110/100/100/100)		
ECM						
Cloak: Cloak (0.25)						

F2534 Cheetah

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
55	50	30	75	90/500	8	180
Weapons				Defense		
L1: Laser (100 (360)) S1: Launcher (5 Splinter (360)) N1: Launcher (5 NanoMissile (360)) R1: Launcher (20 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (100/100/100/100)		
ECM						
Cloak: Cloak (0.25)						

F2533 Lion

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance			
45	40	20	90	80/500	6	270			
Weapons				Defense					
L1: Laser (50 (360)) S1: Launcher (4 Splinter (360)) S2: Launcher (3 Splinter (315, 45)) N1: Launcher (7 NanoMissile (315, 45)) R1: Launcher (20 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (100/100/100/100)					
ECM									
Cloak: Cloak (0.25)									

H2545 Cairo

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	35	25	100	125/500	8	180
Weapons				Defense		
L1: Laser (180 (360)) S1: Launcher (4 Splinter (270, 90)) R1: Launcher (10 Rocket (360)) R2: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) E1: Launcher (5 EMPMissile (360)) G: Gravscan (100)				Shield (150/100/130/100)		

H2552 Babylon

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	35	20	110	100/500	7	210

Weapons	Defense
L1: Laser (180 (270, 90)) S1: Launcher (10 Splinter (90, 270)) R1: Launcher (15 Rocket (360)) N1: Launcher (10 NanocyteMine (360)) G: Gravscan (100)	Shield (150/130/140/130)
ECM	
C1: Cloak (0.2)	

H2535 Rome

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	30	15	150	60/500	5	300
Weapons				Defense		
SS1: Launcher (10 Splinter (0, 180)) SS2: Launcher (10 Splinter (0, 180)) SP1: Launcher (10 Splinter (180, 0)) SP2: Launcher (10 Splinter (180, 0)) R1: Launcher (10 Rocket (360)) N1: Launcher (10 NanoMissile (315, 45)) G: Gravscan (100)				Shield (100/100/100/100)		
ECM						
Cloak: Cloak (0.2)						

H2527 Athens

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	40	20	100	90/500	8	210
Weapons				Defense		
L1: Laser (180 (360)) L2: Laser (180 (360)) R1: Launcher (8 Rocket (360)) R2: Launcher (8 Rocket (360)) G: Gravscan (100)				Shield (100/100/100/100)		
ECM						
Cloak: Cloak (0.2)						

I2544 Hive

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	25	15	150	70/500	6	270
Weapons				Defense		
L1: Laser (100 (360)) SA1: Launcher (5 Splinter (360)) SF1: Launcher (5 Splinter (270, 90))				Shield (140/140/140/140)		

SF2: Launcher (5 Splinter (270, 90)) R1: Launcher (10 Rocket (360)) N1: Launcher (5 NanoMissile (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)	
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I2552 Swarm

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
25	20	10	180	80/500	6	240
Weapons				Defense		
L1: Laser (110 (360)) S1: Launcher (10 Splinter (360)) SF1: Launcher (10 Splinter (315, 45)) SF2: Launcher (5 Splinter (270, 90)) R1: Launcher (10 Rocket (360)) RF1: Launcher (10 Rocket (315, 45)) N1: Launcher (10 NanoMissile (315, 45)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (160/150/150/150)		

I2526 Colony

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	30	15	150	90/500	6	270
Weapons				Defense		
L1: Laser (150 (315, 45)) S1: Launcher (6 Splinter (315, 45)) S2: Launcher (6 Splinter (315, 45)) S3: Launcher (6 Splinter (315, 45)) S4: Launcher (6 Splinter (315, 45)) N1: Launcher (12 NanoMissile (315, 45)) R1: Launcher (20 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (260/125/50/125)		

R2545 Komodo

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
35	30	15	110	90/500	7	210
Weapons				Defense		
L1: Laser (130 (360)) S1: Launcher (10 Splinter (360)) N1: Launcher (10 NanoMissile (360)) R1: Launcher (8 Rocket (360)) R2: Launcher (8 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (120/100/100/100)		

R2525 Viper

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	35	20	80	90/500	7	180
Weapons				Defense		
L1: Laser (120 (360)) S1: Launcher (5 Splinter (360)) N1: Launcher (5 NanoMissile (360)) R1: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (100/100/100/100)		
ECM						
Cloak: Cloak (0.2)						

R2531 Dragon

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	25	15	150	80/500	5	270
Weapons				Defense		
S1: Launcher (5 Splinter (360)) S2: Launcher (5 Splinter (360)) S3: Launcher (5 Splinter (360)) N1: Launcher (10 NanoMissile (360)) R1: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (120/120/120/120)		
ECM						
Cloak: Cloak (0.2)						

R2551 Cobra

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
35	30	15	105	90/500	7	210
Weapons				Defense		
L1: Laser (80 (360)) S1: Launcher (4 Splinter (360)) S2: Launcher (4 Splinter (360)) N1: Launcher (5 NanoMissile (360)) R1: Launcher (10 Rocket (360)) M1: Launcher (10 SplinterMine (360)) G: Gravscan (100)				Shield (120/100/100/100)		

Starbases

Starbases can not move, but they have powerful weapons and defense, and can replenish ships that fly within the maximum replenish distance and speed, and then give the Replenish command.

SB2531

Replenish distance	Replenish speed	Hull	Battery	Generators	Scan Distance
10	10	250	200/1000	12	390
Weapons			Defense		
L1: Laser (300 (360)) L2: Laser (300 (360)) S1: Launcher (5 Splinter (360)) S2: Launcher (5 Splinter (360)) R1: Launcher (5 Rocket (360)) R2: Launcher (5 Rocket (360)) G: Gravscan (100)			Shield (400/400/400/400)		